5.4: Proteins include a diversity of structures, resulting in a wide range of functions

Nearly every dynamic function of a living being depends on proteins. In fact, the importance of proteins is underscored by their name, which comes from the Greek word *proteios*, meaning “first,” or “primary.” Proteins account for more than 50% of the dry mass of most cells, and they are instrumental in almost everything organisms do. Some proteins speed up chemical reactions, while others play a role in defense, storage, transport, cellular communication, movement, or structural support. **Figure 5.13** shows examples of proteins with these functions, which you’ll learn more about in later chapters.

Figure 5.13

An overview of protein functions.

An overview of protein functions.

Storage proteins. Function: Storage of amino acids. Examples: Casein, the protein of milk, is the major source of amino acids for baby mammals. Plants have storage proteins in their seeds. Ovalbumin is the protein of egg white, used as an amino acid source for the developing embryo.

Hormonal proteins. Vessel with high blood sugar levels causes insulin to be secreted, resulting in vessel with normal blood sugar levels. Function: Coordination of an organism‘s activities. Example: Insulin, a hormone secreted by the pancreas, causes other tissues to take up glucose, thus regulating blood sugar concentration.

Contractile and motor proteins. Function: Movement. Examples: Motor proteins are responsible for the undulations of cilia and flagella. Actin and myosin proteins are responsible for the contraction of muscles.

Defensive proteins. Function: Protection against disease. Example: Antibodies inactivate and help destroy viruses and bacteria.

Transport proteins. A transport protein imbedded in the cell membrane allows for the passage of a substance into the cell. Function: Transport of substances. Examples: Hemoglobin, the iron-containing protein of vertebrate blood, transports oxygen from the lungs to other parts of the body. Other proteins transport molecules across membranes, as shown here.

Receptor proteins. A nerve cell releases signaling molecules that bind to receptors on the surface of another nerve cell. Function: Response of cell to chemical stimuli. Example: Receptors built into the membrane of a nerve cell detect signaling molecules released by other nerve cells.

Structural proteins. Function: Support. Examples: Keratin is the protein of hair, horns, feathers, and other skin appendages. Insects and spiders use silk fibers to make their cocoons and webs, respectively. Collagen and elastin proteins provide a fibrous framework in animal connective tissues.

Animation

Protein Functions

image

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Life would not be possible without enzymes, most of which are proteins. Enzymatic proteins regulate metabolism by acting as **catalysts**, chemical agents that selectively speed up chemical reactions without being consumed in the reaction. Because an enzyme can perform its function over and over again, these molecules can be thought of as workhorses that keep cells running by carrying out the processes of life.

A human has tens of thousands of different proteins, each with a specific structure and function; proteins, in fact, are the most structurally sophisticated molecules known. Consistent with their diverse functions, they vary extensively in structure, each type of protein having a unique three-dimensional shape.

Proteins are all constructed from the same set of 20 amino acids, linked in unbranched polymers. The bond between amino acids is called a *peptide bond*, so a polymer of amino acids is called a **polypeptide**. A **protein** is a biologically functional molecule made up of one or more polypeptides, each folded and coiled into a specific three-dimensional structure.

Amino Acids (Monomers)

All amino acids share a common structure. An **amino acid** is an organic molecule with both an amino group and a carboxyl group (see Figure 4.9); the small figure shows the general formula for an amino acid. At the center of the amino acid is an asymmetric carbon atom called the *alpha* α *carbon*. Its four different partners are an amino group, a carboxyl group, a hydrogen atom, and a variable group symbolized by R. The R group, also called the side chain, differs with each amino acid. The R group may be as simple as a hydrogen atom, or it may be a carbon skeleton with various functional groups attached. The physical and chemical properties of the side chain determine the unique characteristics of a particular amino acid, thus affecting its functional role in a polypeptide.

Structure of an amino acid.

A central carbon (alpha carbon) is linked to an amino group (N H 2), a carboxyl group (C double bonded to O and single bond to O H), an H, and an R group (side chain).

Figure 5.14 shows the 20 amino acids that cells use to build their thousands of proteins. Here the amino groups and carboxyl groups are all depicted in ionized form, the way they usually exist at the pH found in a cell. The amino acids are grouped according to the properties of their side chains. One group consists of amino acids with nonpolar side chains, which are hydrophobic. Another group consists of amino acids with polar side chains, which are hydrophilic. Acidic amino acids have side chains that are generally negative in charge due to the presence of a carboxyl group, which is usually dissociated (ionized) at cellular pH. Basic amino acids have amino groups in their side chains that are generally positive in charge. (The terms *acidic* and *basic* in this context refer only to groups in the side chains because *all* amino acids—as monomers—have carboxyl groups and amino groups.) Because they are charged, acidic and basic side chains are also hydrophilic.

Figure 5.14

The 20 amino acids of proteins.

Detailed structures and abbreviations for the 20 natural amino acids.

The 20 natural amino acids are grouped here according to the properties of their side chains (R groups) and shown in their prevailing form at pH 7.2, the pH within a cell. The three-letter and one letter abbreviations for each amino acid are given in parenthesis. Each of the amino acids has the same general structure, namely, H3N+–CHR–COO-. The R group is the only part that differs. Throughout this description, groups of atoms that branch off the main chain of the R group are set off with parenthesis in the formula. All of the amino acids used in proteins are L enantiomers (see Figure 4.7c).

a. Nonpolar (Hydrophobic) Side Chains.

- glycine (Gly or G): R = single bond H.
- alanine (Ala or A): R = single bond C H 3 (methyl).
- valine (Val or V): R = single bond C H, C H 3, 2, (isopropyl).
- leucine (Leu or L): R = single bond C H 2. single bond, C H, C H 3, 2 (isobutyl).
- isoleucine (Ile or I): R = single bond C H, C H 3, single bond, C H 2, single bond, C H 3 (sec-butyl).
- methionine (Met or M): R = single bond C H 2, single bond, C H 2, single bond, S, single bond, C H 3.
- phenylalanine (Phe or F): R = single bond, C H 2, single bond, C 6 H 5 (benzyl). This side chain contains an aromatic ring.

- tryptophan (Trp or W): R = single bond C H 2, single bond, indole. The R group for this amino acid is relatively complex: Its CH2 group is bonded to the number 3 atom of an indole molecule, which consists of fused five- and six-membered rings. To draw indole, start with a five-membered ring containing an NH group at position 1. Draw double bonds between carbons 2 and 3 and between carbons 4 and 5. Carbon 3 is bonded to the CH2 leading to the backbone of the amino acid. Complete the indole molecule, using carbons 4 and 5 as the basis for forming the six-membered ring. Make the six-membered ring aromatic by including two more double bonds.
- proline (Pro or P): R = single bond, C H 2, single bond, CH 2, single bond, CH 2, single bond. Proline is unique, as it is the only amino acid where the R group wraps around to forms a ring with the backbone (specifically, the amino group) of the amino acid. This makes the backbone have the slightly different form R N H 2 plus C H R, single bond, C O 2 plus, where both R's refers to the same three carbon bridge).

b. Polar (Hydrophilic) Side Chains:

- serine (Ser or S): R = single bond C H 2, single bond, O H.
- threonine (Thr or T): R = single bond, C H, O H, single bond, C H 3.
- cysteine (Cys or C): R = single bond C H2, single bond, S H.
- tyrosine (Tyr or Y): R = single bond C H2, single bond, C 6 H 4, single bond, O H (Tyrosine contains an aromatic ring. The single bond O H is para to the amino acid backbone.)
- asparagine (Asn or N): R = single bond C H2, single bond, C O, single bond, N H 2 (contains a carbonyl, amide).
- glutamine (Gln or Q): R = single bond, C H 2, single bond, C H 2, single bond, C O, single bond, N H 2 (contains a carbonyl, amide.

c. Electrically charged side chains (hydrophilic), Acidic, negatively charged:

- aspartic acid (Asp or D): R = single bond, C H 2, single bond, C O O- (contains a carboxylate ion in neutral solutions).
- glutamic acid (Glu or E): R = single bond, C H 2, single bond, C H 2, single bond, C O O- (contains a carboxylate ion in neutral solutions).

d. Electrically charged side chains (hydrophilic), Basic, Positively Charged:

- lysine (Lys or K): R = –single bond, C H 2, single bond, C H 2, single bond, C H 2, single bond, C H 2, single bond, N H 3 plus.

- arginine (Arg or R): R = single bond, C H 2, single bond, C C H 2, single bond, C H 2, single bond, N H, single bond, C, N H 2, plus, single bond, N H 2. The NH 2 plus is double-bonded to the carbon.
- histidine (His or H): R = single bond, C H 2, single bond, imidazole. Histidine has a relatively complex structure. To draw it, begin with the structure of imidazole, a five-membered ring containing a nitrogen atom at both positions 1 and 3. Add a hydrogen atom to both nitrogen atoms, and a plus charge to nitrogen 3. Draw a double bond between carbon 2 and nitrogen 3, and another double bond between carbons 4 and 5. Carbon 5 bonds to the CH2 leading back to the amino acid backbone.

At the top right are the non-ionized and ionized forms of a generic amino acid. The specific amino acids are shown below in their ionized form, prevalent at the pH within a cell (pH 7.2). The amino acids are grouped by properties of their side chains. The three-letter and one-letter abbreviations for the amino acids are in parentheses. All of the amino acids used in proteins are L enantiomers (see Figure 4.7c).

Polypeptides (Amino Acid Polymers)

Now that we have examined amino acids, let's see how they are linked to form polymers **(Figure 5.15)**. When two amino acids are positioned so that the carboxyl group of one is adjacent to the amino group of the other, they can become joined by a dehydration reaction, with the removal of a water molecule. The resulting covalent bond is called a **peptide bond**. Repeated over and over, this process yields a polypeptide, a polymer of many amino acids linked by peptide bonds. You'll learn more about how cells synthesize polypeptides in Concept 17.4.

Figure 5.15

Making a polypeptide chain.

The structure and synthesis of a polypeptide chain.

The polypeptide backbone consists of a series of amino acids held together with peptide bonds. The left end of the backbone as drawn is called the amino end (N-terminus). Beginning here, the backbone is series of repeating N-C-C amino acid units. Each amino acid has a side chain on its middle (alpha) carbon. The carboxyl carbon of each amino acid forms a peptide bond with the nitrogen of the next amino acid. The right-hand end of the polypeptide is called the carboxyl end (C-terminus). New amino acids are added one at a time to the carboxyl end of the peptide. With each new amino acid, a dehydration reaction makes a water molecule from the O H of the peptide's carboxyl end and an H atom from the amino group of the new amino acid. A new peptide bond is also formed between the carboxyl and amino groups.

Peptide bonds are formed by dehydration reactions, which link the carboxyl group of one amino acid to the amino group of the next. The peptide bonds are formed one at a time, starting with the amino acid at the amino end (N-terminus). The polypeptide has a repetitive backbone (purple) to which the amino acid side chains (yellow and green) are attached.

DRAW IT *Label the three amino acids in the upper part of the figure using three-letter and one-letter codes. Circle and label the carboxyl and amino groups that will form the new peptide bond.*

*For suggested answer, see**Appendix A*.

The repeating sequence of atoms highlighted in purple in Figure 5.15 is called the *polypeptide backbone*. Extending from this backbone are the different side chains (R groups) of the amino acids. Polypeptides range in length from a few amino acids to 1,000 or more. Each specific polypeptide has a unique linear sequence of amino acids. Note that one end of the polypeptide chain has a free amino group (the N-terminus of the polypeptide), while the opposite end has a free carboxyl group (the C-terminus). The chemical nature of the molecule as a whole is determined by the kind and sequence of the side chains, which determine how a polypeptide folds and thus its final shape and chemical characteristics. The immense variety of polypeptides in nature illustrates an important concept introduced earlier—that cells can make many different polymers by linking a limited set of monomers into diverse sequences.

Protein Structure and Function

The specific activities of proteins result from their intricate three-dimensional architecture, the simplest level of which is the sequence of their amino acids. What can the amino acid sequence of a polypeptide tell us about the three-dimensional structure (commonly referred to simply as the “structure”) of the protein and its function? The term *polypeptide* is not synonymous with the term *protein*. Even for a protein consisting of a single polypeptide, the relationship is somewhat analogous to that between a long strand of yarn and a sweater of particular size and shape that can be knitted from the yarn. A functional protein is not *just* a polypeptide chain, but one or more polypeptides precisely twisted, folded, and coiled into a molecule of unique shape, which can be shown in several different types of models **(Figure 5.16)**. And it is the amino acid sequence of each polypeptide that determines what three-dimensional structure the protein will have under normal cellular conditions.

Advanced placement edition

Apply concepts and connect across Big Ideas using visual representations.

What determines the properties of a protein? What structural properties are critical for membrane-bound transport proteins? How is a change in a protein and gene expression related to a change in DNA sequence?

Figure 5.16

Visualizing Proteins

Visualizing proteins.

Proteins can be represented in different ways, depending on the goal of the illustration.

Structural models: Using data from structural studies of proteins, computers can generate various types of models. Each model emphasizes a different aspect of the protein's structure, but no model can show what a protein actually looks like. These three models depict lysozyme, a protein in tears and saliva that helps prevent infection by binding to target molecules on bacteria. From left to right, these models are space-filling, ribbon, and wireframe.

- Space-filling model: Shows all the atoms of the protein (except hydrogen), emphasizing the overall globular shape. The atoms are color-coded: gray = carbon, red = oxygen, blue = nitrogen, and yellow = sulfur. Groove is an indent in the model.
- Ribbon model: Shows only the backbone of the polypeptide, emphasizing how it folds and coils to form a 3-D shape, in this case stabilized by disulfide bridges (yellow lines). Groove is shown as an indent in the folded ribbon structure.
- Wireframe model (blue): Shows the backbone of the polypeptide chain with side chains (R groups) extending from it (see Figure 5.15). A ribbon model (purple) is superimposed on the wireframe model. Target molecule (on the bacterial cell surface) bound to lysozyme, shown joining with "groove" in model.

Simplified diagrams: It isn't always necessary to use a detailed computer J17model; simplified diagrams are useful when the focus of the figure is on the function of the protein, not the structure. In this diagram of the protein rhodopsin, a simple transparent shape is drawn around the contours of a ribbon model, showing the overall shape of the molecule as well as some internal details. When structural details are not needed, a solid shape can be used to represent a protein. Image shows protein as a purple oval shaded to look three-dimensional. A simple shape is used here to represent a generic enzyme because the diagram focuses on enzyme action in general. Image shows two enzymes, both shown as purple ovals. The first enzyme is shown with two green hexagonal food molecules which are linked to each other. An arrow flows to the second enzyme which has broken the bonds between the food molecules, and the molecules are shown moving away from the enzyme. Sometimes a protein is represented simply as a dot, as shown here for insulin. Image shows insulin-producing cell in pancreas. Cell is an orange oval with four small green dots on its left side representing insulin.

Instructors: The tutorial "Molecular Model: Lysozyme," in which students rotate 3-D models of lysozyme, can be assigned in **Mastering Biology**.

Instructors: Additional questions related to this Visualizing Figure can be assigned in **Mastering Biology**.

When a cell synthesizes a polypeptide, the chain may fold spontaneously, assuming the functional structure for that protein. This folding is driven and reinforced by the formation of various bonds between parts of the chain, which in turn depends on the sequence of amino acids. Many proteins are roughly spherical (*globular proteins*), while others are shaped like long fibers (*fibrous proteins*). Even within these broad categories, countless variations exist.

A protein's specific structure determines how it works. In almost every case, the function of a protein depends on its ability to recognize and bind to some other molecule. In an especially striking example of the marriage of form and function, **Figure 5.17** shows the exact match of shape between an antibody (a protein in the body) and the particular foreign substance on a flu virus that the antibody binds to and marks for destruction. Also, you may recall another example of molecules with matching shapes from Concept 2.3: endorphin molecules (produced by the body) and morphine molecules (a manufactured drug), both of which fit into receptor proteins on the surface of brain cells in humans, producing euphoria and relieving pain. Morphine, heroin, and other opiate drugs are able to mimic endorphins because they all have a shape similar to that of endorphins and can thus fit into and bind to endorphin receptors in the brain. This fit is very specific, something like a handshake (see Figure 2.16). The endorphin receptor, like other receptor molecules, is a protein. The function of a protein—for instance, the ability of a receptor protein to bind to a particular pain-relieving signaling molecule—is an emergent property resulting from exquisite molecular order.

Figure 5.17

Complementarity of shape between two protein surfaces.

Different parts of an antibody protein and a protein from the flu virus are colored. The protein and virus have complementary shapes in parts.

A technique called X-ray crystallography was used to generate a computer model of an antibody protein (blue and orange, left) bound to a flu virus protein (yellow and green, right). This is a wireframe model modified by adding an "electron density map" in the region where the two proteins meet. Computer software was then used to back the images away from each other slightly.

VISUAL SKILLS *What do these computer models allow you to see about the two proteins?*

*For suggested answer, see**Appendix A.*

Four Levels of Protein Structure

In spite of their great diversity, proteins share three superimposed levels of structure, known as **primary structure**, **secondary structure**, and **tertiary structure**. A fourth level, quaternary structure, arises when a protein consists of two or more polypeptide chains. **Figure 5.18** describes these four levels of protein structure. Be sure to study this figure thoroughly before going on to the next section.

Animation

Protein Structure

image

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Figure 5.18

Exploring Levels of Protein Structure

Exploring the levels of protein structure.

The primary structure of a protein is its sequence of amino acids. As an example, let's consider transthyretin, a globular blood protein that transports vitamin A and one of the thyroid hormones throughout the body. Transthyretin is made up of four identical polypeptide chains, each composed of 127 amino acids. Shown here is one of these chains unraveled for a closer look at its primary structure. Each of the 127 positions along the chain is occupied by one of the 20 amino acids, indicated here by its three-letter abbreviation. The primary structure is like the order of letters in a very long word. If left to chance, there would be 20127 different ways of making a polypeptide chain 127 amino acids long. However, the precise primary structure of a protein is determined not by the random linking of amino acids, but by inherited genetic information. The primary structure in turn dictates secondary and tertiary structure, due to the chemical nature of the backbone and the side chains (R groups) of the amino acids along the polypeptide. Each amino acid is made up of a central carbon attached to an amino group, a carboxyl group, an H and a side group (one of 20 amino acids).

Secondary structure: Regions stabilized by hydrogen bonds between atoms of the polypeptide backbone. Includes alpha helices and beta pleated sheets with structures supported by hydrogen bonds. Most proteins have segments of their polypeptide chains repeatedly coiled or folded in patterns that contribute to the

protein's overall shape. These coils and folds, collectively referred to as secondary structure, are the result of hydrogen bonds between the repeating constituents of the polypeptide backbone (not the amino acid side chains). Within the backbone, the oxygen atoms have a partial negative charge, and the hydrogen atoms attached to the nitrogens have a partial positive charge (see Figure 2.14); therefore, hydrogen bonds can form between these atoms. Individually, these hydrogen bonds are weak, but because they are repeated many times over a relatively long region of the polypeptide chain, they can support a particular shape for that part of the protein. One such secondary structure is the alpha helix, a delicate coil held together by hydrogen bonding between every fourth amino acid, as shown above. Although each transthyretin polypeptide has only one alpha helix region (see the Tertiary Structure section), other globular proteins have multiple stretches of alpha helix separated by nonhelical regions (see hemoglobin in the Quaternary Structure section). Some fibrous proteins, such as alpha-keratin, the structural protein of hair, have the alpha helix formation over most of their length. The other main type of secondary structure is the beta pleated sheet. As shown above, in this structure two or more segments of the polypeptide chain lying side by side (called beta strands) are connected by hydrogen bonds between parts of the two parallel segments of polypeptide backbone. Beta pleated sheets make up the core of many globular proteins, as is the case for transthyretin (see Tertiary Structure), and dominate some fibrous proteins, including the silk protein of a spider's web. The teamwork of so many hydrogen bonds makes each spider silk fiber stronger than a steel strand of the same weight. Spiders secrete silk fibers made of a structural protein containing beta pleated sheets, which allow the spider web to stretch and recoil.

Tertiary structure: three-dimensional shape stabilized by interactions between side chains. Diagram shows ribbon model of transthyretin polypeptide, which includes regions of both alpha helices and beta pleated sheets. Superimposed on the patterns of secondary structure is a protein's tertiary structure, shown here in a ribbon model of the transthyretin polypeptide. While secondary structure involves interactions between backbone constituents, tertiary structure is the overall shape of a polypeptide resulting from interactions between the side chains (R groups) of the various amino acids. One type of interaction that contributes to tertiary structure is called, somewhat misleadingly, a hydrophobic interaction. As a polypeptide folds into its functional shape, amino acids with hydrophobic (nonpolar) side chains usually end up in clusters at the core of the protein, out of contact with water. Thus, a "hydrophobic interaction" is actually caused by the exclusion of nonpolar substances by water molecules. Once nonpolar amino acid side chains are close together, van der Waals interactions help hold them together. Meanwhile, hydrogen bonds between polar side chains and ionic bonds between positively and negatively charged side chains also help stabilize tertiary structure. These are all weak interactions in the aqueous cellular environment, but their cumulative effect helps give the protein a unique shape. Covalent bonds called disulfide bridges may further reinforce the shape of a protein. Disulfide bridges form where two cysteine monomers, which have sulfhydryl groups (S H) on their side chains (see Figure 4.9), are brought close together by the folding of the protein. The sulfur of

one cysteine bonds to the sulfur of the second, and the disulfide bridge (single bond, S, single bond, S, single bond) rivets parts of the protein together (see yellow lines in Figure 5.16 ribbon model). All of these different kinds of interactions can contribute to the tertiary structure of a protein.

Quaternary Structure: Association of two or more polypeptides (some proteins only). Transthyretin protein is made up of four identical polypeptides, each with secondary, tertiary and primary structure. Some proteins consist of two or more polypeptide chains aggregated into one functional macromolecule. Quaternary structure is the overall protein structure that results from the aggregation of these polypeptide subunits. For example, shown above is the complete globular transthyretin protein, made up of its four polypeptides. Another example is collagen, which is a fibrous protein that has three identical helical polypeptides intertwined into a larger triple helix, giving the long fibers great strength. This suits collagen fibers to their function as the girders of connective tissue in skin, bone, tendons, ligaments, and other body parts. (Collagen accounts for 40% of the protein in a human body.)

Animation

Proteins

image

Play

image

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MP3 Tutor

Protein structure and Function

Hey there, and welcome to another MP3 Tutor session. This is Eric Simon and I'll guide you through today's topic: proteins. In your text and in other MP3 sessions, you've been exposed to one of the major themes of biology: the relationship between structure and function. In biology, we say that form follows function. That is, the structure of an object reflects the job done. For example, the form of a chair must reflect its function. Chairs can come in many different shapes and sizes, but they all must have a flat part to sit on and a stable part to meet the floor. Throughout multiple levels of biological organization, exploring the theme of structure and function can help you to develop a deeper understanding of how life works. In this MP3 tutor session, we will explore how proteins are put together and relate that structure to a few specific examples of protein function. Sound good? Let's get started. When I hear the word "protein" the first thing that usually pops in my mind is food: meat, dairy products, eggs, that sort of thing. Of course, I think about food a lot.

However, even when my appetite is saying "salad," I always make sure that my salad has ample protein in the form of nuts, seeds, and beans. Proteins in food provide the raw material to build the bodily proteins essential for your good health. Proteins have the most diverse and complex architecture of any of the biological macromolecules. You have tens of thousands of different proteins in your body that have diverse structures and perform diverse jobs. Proteins are the worker bees of your body. They take part in nearly every activity of an organism. Proteins play a role in structural support (as in cartilage), transport (such as hemoglobin), hormones (like insulin), immunity (your antibodies), movement (as with muscle proteins), and as enzymes that keep thousands of different chemical reactions running inside all cells. In each of these examples, the shape or structure of a protein is the key to understanding how it works. How can the shape of a protein be so important? Let's try an analogy. Pliers are one of the most commonly used tools for extending the power and grip of your hand. Most people have used pliers to accomplish a specific task, such as cracking a nut, removing a bolt, bending wire, cutting wire, and so on. In each case, the moveable jaws of the pliers are shaped for specific tasks. Using them for a different duty sometimes leads to frustration. For example, you can't remove a bolt with a pair of needle-nosed pliers. Shape is important to a tool's function, and the same is true for proteins. Each of the thousands of different kinds of proteins in your body has its own unique shape. Describing so many different proteins, with so many different shapes, sounds like a daunting task. Luckily, it's not as difficult as it sounds. Let's explore how proteins are put together. When describing the structure of proteins, biologists think about four distinct levels of protein structure. These four levels are known as primary, secondary, tertiary, and quaternary. I always have trouble pronouncing these words so sometimes I simply refer to these levels as the first, second, third or fourth level of protein structure. The first or primary level of protein structure is simply the linear sequence of amino acids that are strung together to make the protein chain. Every protein has a unique amino acid sequence. Just as you can build thousands of words using just 26 letters of the English alphabet, a huge variety of proteins can be constructed from 20 different amino acids. Just like English letters have different properties-such as consonants vs. vowels-so do amino acids. The properties of an individual amino acid lay in its side chain that branches off of the carbon backbone. The side chain can also be called the R group or radical group. Let me use another analogy. Different railroad cars (such as tanker cars, flatbed cars, or refrigerator cars) can be built on a standard railroad car chassis so that cars can be easily linked together to form a long train. Likewise, a common carbon backbone is found in all amino acids. This common backbone permits amino acids to be linked together in long chains. Unlike a train, however, the chain of amino acids folds and twists to form a complex three dimensional shape. The shape eventually taken by a chain of amino acids is directly tied to the side groups that protrude from the amino acid chain. If even one amino acid is out of order it can make the entire protein non-functioning. However, let me point out that some amino acids can be substituted for others without altering the function of a protein. Let's go back to the word analogy for a second. If you change just one letter in some words you can completely change the meaning. For example, changing one letter in the word "ship" can produce a complete different word like "shop"-very different

meanings, right? Other words, however, are not as susceptible to single changes. If you change a letter around, they may be misspelled, but their meaning may not be changed. For example, if you switch around the "e" and the "i" in the word "protein" you'd have no trouble understanding the meaning of the word. If the amino acid sequence of a protein is changed, most changes result in defective proteins but some do not. To recap, the primary structure of a protein is simply the list of amino acids in their proper order. To understand the secondary level of protein structure, imagine that you are an alien seeing a roomful of human beings for the first time. The first thing you might notice is that every person has a unique shape. But you would quickly start to notice similarities in the shapes. Every person, for example, has a rectangular body with a spherical blob at the top and four long limbs sticking out. If you look at many proteins, you will similarly notice certain common structural elements. The two most common are called an alpha helix and a beta sheet. An alpha helix is a region of a protein that is twisted into a tight spiral. The coils of the helix are held together by hydrogen bonds between nearby amino acids. Other regions of a protein fold up like a paper fan. This type of structure is referred to as a beta sheet. The alpha helices and the beta sheets together comprise the secondary structure of a protein. A given protein might have three alpha helices and four beta sheets, while another has eight and two, and so on. Most proteins have at least some of each. Once all the helices and sheets are coiled, folded, and locked into place, the polypeptide takes on a distinct three-dimensional shape. This is the third or tertiary level of protein structure: the overall conformation or three-dimensional shape of a protein. To think about protein shapes, imagine that I have a big barrel of clay and I ask everyone in the class to grab a lump and put it on their desk. At first glance, all the lumps look similar; they are all blobby lumps of clay. But closer observation reveals that every lump of clay is unique. Some are taller or fatter, some have bumps and grooves that others lack. Similarly, while most proteins look basically like round-ish blobs, each one has small differences in their structure. And these differences in shape are vital to the job a protein performs. That is, form follows function. Change a protein's shape in the slightest (by, say, cutting off a lump) and the protein may not be able to perform its job within the cell. The fourth level of protein structure describes interactions among different polypeptides. Some complex proteins are composed of more than one polypeptide chain that bond together to form a sort of super protein. For example, the blood protein hemoglobin is composed of four different polypeptide chains that interlock. Another example is the molecular machine known as DNA polymerase, the enzyme that builds new DNA molecules during DNA replication. DNA polymerase is actually a large complex made up of a number of different polypeptide chains joined together. Not all proteins have quaternary structure, but the big ones usually do. As a summary, let's review the four levels of protein structure: The first level is the sequence of amino acids. The second level is the formation of alpha helices and beta sheets. The third level is the folding of a polypeptide into its final three dimensional shape. The fourth level is the joining of more than one chain to form a larger complex. I hope this gives you a leg up on this topic. If you need more help, please refer to your book, CD-ROM, or course website. Good luck with your studies.

Sickle-Cell Disease: A Change in Primary Structure

Even a slight change in primary structure can affect a protein's shape and ability to function. For instance, **sickle-cell disease**, an inherited blood disorder, is caused by the substitution of one amino acid (valine) for the normal one (glutamic acid) at the position of the sixth amino acid in the primary structure of hemoglobin, the protein that carries oxygen in red blood cells. Normal red blood cells are disk-shaped, but in sickle-cell disease, the abnormal hemoglobin molecules tend to aggregate into chains, deforming some of the cells into a sickle shape **(Figure 5.19)**. A person with the disease has periodic "sickle-cell crises" when the angular cells clog tiny blood vessels, impeding blood flow. The toll taken on such patients is a dramatic example of how a simple change in protein structure can have devastating effects on protein function.

Figure 5.19

A single amino acid substitution in a protein causes sickle-cell disease.

Normal and sickle-cell hemoglobin is compared.

The table is as follows.

Hemoglobin Type	Primary Structure	Secondary and Tertiary Structures	Quaternary Structure	Function	Red Blood Cell Shape
Normal hemoglobin	Sequence of amino acids: Val, His, Leu, Thr, Pro, Glu, Glu.	Normal beta subunit A spherical, blue colored subunit is depicted.	Normal alpha subunits and 2 normal beta Two spherical, blue-colored beta subunits are superimposed on purple, smaller alpha subunits.	Normal hemoglobin proteins do not associate with one another; each carries oxygen. Small images of hemoglobin with blue and purple subunits are depicted.	Normal red blood cells are full of individual hemoglobin molecules. An image of a normal red blood cell shows a disc shape (scale: 5 micrometers).
Sickle-cell hemoglobin	Sequence of amino acids: Val, His, Leu, Thr, Pro, Val,	Sickle-cell beta subunit is malformed —has a slight	In sickle-cell hemoglobin, both beta subunits have a	Hydrophobic interactions between sickle-cell hemoglobin	Fibers of abnormal hemoglobin deform red

	Glu. The second “Val” was a “Glu” in normal hemoglobin.	“sickle” shape as compared to the normal beta subunit. A spherical, red-colored beta subunit with a small indentation on the left side is shown.	slight sickle shape. Both alpha subunits are normal. A hemoglobin molecule with 2 abnormal beta subunits in red, superimposed on two normal alpha subunits is shown.	proteins lead to their aggregation into a fiber; the capacity to carry oxygen is greatly reduced. A fiber of hemoglobin is shown, which upon magnification shows sickle-cell, abnormal hemoglobin molecules alternately stacked to form a fiber.	blood cells into a sickle shape. Image of a sickle-shaped red blood cell (scale: 5 micrometers).

MAKE CONNECTIONS *Considering the chemical characteristics of the amino acids valine and glutamic acid (see*Figure 5.14*)*, propose a possible explanation for the dramatic effect on protein function that occurs when valine is substituted for glutamic acid.*

*For suggested answer, see**Appendix A*.

HHMI BioInteractive Animation

Sickle-Cell Disease

image

Play

image

**

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Interview

Interview with Linus Pauling: Winner of the Nobel Prize in Chemistry and the Nobel Peace Prize

A photo of Linus Pauling.

What Determines Protein Structure?

You've learned that a unique shape endows each protein with a specific function. But what are the key factors determining protein structure? You already know most of the answer: A polypeptide chain of a given amino acid sequence can be arranged into a three-dimensional shape determined by the interactions responsible for secondary and tertiary structure. This folding normally occurs as the protein is being synthesized in the crowded environment within a cell, aided by other proteins. However, protein structure also depends on the physical and chemical conditions of the protein's environment. If the pH, salt concentration, temperature, or other aspects of its environment are altered, the weak chemical bonds and interactions within a protein may be destroyed, causing the protein to unravel and lose its native shape, a change called **denaturation(Figure 5.20)**. Because it is misshapen, the denatured protein is biologically inactive.

Figure 5.20

Denaturation and renaturation of a protein.

A normal protein goes through denaturation to become a denatured protein. A denatured protein goes through renaturation to become a normal protein.

High temperatures or various chemical treatments will denature a protein, causing it to lose its shape and hence its ability to function. If the denatured protein remains dissolved, it may renature when the chemical and physical aspects of its environment are restored to normal.

See table next tab

Most proteins become denatured if they are transferred from an aqueous environment to a nonpolar solvent, such as ether or chloroform; the polypeptide chain refolds so that its hydrophobic regions face outward toward the solvent. Other denaturation agents include chemicals that disrupt the hydrogen bonds, ionic bonds, and disulfide bridges that maintain a protein's shape. Denaturation can also result from excessive heat, which agitates the polypeptide chain enough to overpower the weak interactions that stabilize the structure. The white of an egg becomes opaque during cooking because the denatured proteins are insoluble and solidify. This also explains why excessively high fevers can be fatal: Proteins in the blood tend to denature at very high body temperatures.

When a protein in a test-tube solution has been denatured by heat or chemicals, it can sometimes return to its functional shape when the denaturing agent is removed. (Sometimes this is not possible: For example, a fried egg will not become liquefied when placed back into the refrigerator!) We can conclude that the information for building specific shape is intrinsic to the protein's primary structure; this is often the case for small proteins. The sequence of amino acids determines the protein's shape—where an **alpha helix can form, where betapleated sheets** can exist, where disulfide bridges are located, where ionic bonds can form, and so on. But how does protein folding occur in the cell?

Protein Folding in the Cell

Biochemists now know the amino acid sequence for over a hundred million proteins, with several million added each month, and the three-dimensional shape for tens of thousands. Researchers have tried to correlate the primary structure of many proteins with their three-dimensional structure to discover the rules of protein folding. Unfortunately, however, the protein-folding process is not that simple. Most proteins probably go through several intermediate structures on their way to a stable shape, and looking at the mature structure does not reveal the stages of folding required to achieve that form. However, biochemists have developed methods for tracking a protein through such stages and learning more about this important process.

Misfolding of polypeptides in cells is a serious problem that has come under increasing scrutiny by medical researchers. Many diseases—such as cystic fibrosis, Alzheimer's, Parkinson's, and mad cow disease—are associated with an accumulation of misfolded proteins. In fact, misfolded versions of the transthyretin protein featured in Figure 5.18 have been implicated in several diseases, including one form of senile dementia.

Even when scientists have a correctly folded protein in hand, determining its exact three-dimensional structure is not simple, for a single protein has thousands of atoms. The method most commonly used to determine the 3-D structure of a protein is **X-ray crystallography**, which depends on the diffraction of an X-ray beam by the atoms of a crystallized molecule. Using this technique, scientists can build a 3-D model that shows the exact position of every atom in a protein molecule **(Figure 5.21)**. Nuclear magnetic resonance (NMR) spectroscopy, cryo-electron microscopy (cryo-EM; see Concept 6.1) and bioinformatics (see Concept 5.6) are complementary approaches to understanding protein structure and function.

Figure 5.21

Research Method: X-Ray Crystallography

Application

Scientists use X-ray crystallography to determine the three-dimensional 3-D structure of macromolecules such as nucleic acids and proteins.

Technique

Researchers aim an X-ray beam through a crystallized protein or nucleic acid. The atoms of the crystal diffract (bend) the X-rays into an orderly array that a digital detector records as a pattern of spots called an X-ray diffraction pattern, an example of which is shown here.

X-ray crystallography.

Application: Scientists use X-ray crystallography to determine the three-dimensional structure of macromolecules such as nucleic acids and proteins.

Technique. Researchers aim an x-ray beam through crystalized protein or nucleic acid. The atoms of the crystal diffract (bend) the x-rays into an orderly array that a digital detector records as a pattern of spots called x-ray diffraction pattern, an example of which is given as the following diagram. An x-ray source sends a beam through a crystal. The beam scatters and hits a digital detector. A pattern emerges.

Results. Using data from x-ray diffraction patterns and the sequence of monomers determined by chemical methods, researchers can build a 3-D computer model of the macromolecule being studied, such as the four-subunit protein transthyretin.

Results

Using data from X-ray diffraction patterns and the sequence of monomers determined by chemical methods, researchers can build a 3-D computer model of the macromolecule being studied, such as the four-subunit protein transthyretin (see Figure 5.18) shown here.

A 3 D model of transthyretin, with 4 subunits, each colored differently.

The structure of some proteins is difficult to determine for a simple reason: A growing body of biochemical research has revealed that a significant number of proteins, or regions of proteins, do not have a distinct 3-D structure until they interact with a target protein or other molecule. Their flexibility and indefinite structure are important for their function, which may require binding with different targets at different times. These proteins, which may account for 20–30% of mammalian proteins, are called *intrinsically disordered proteins* and are the focus of current research.